

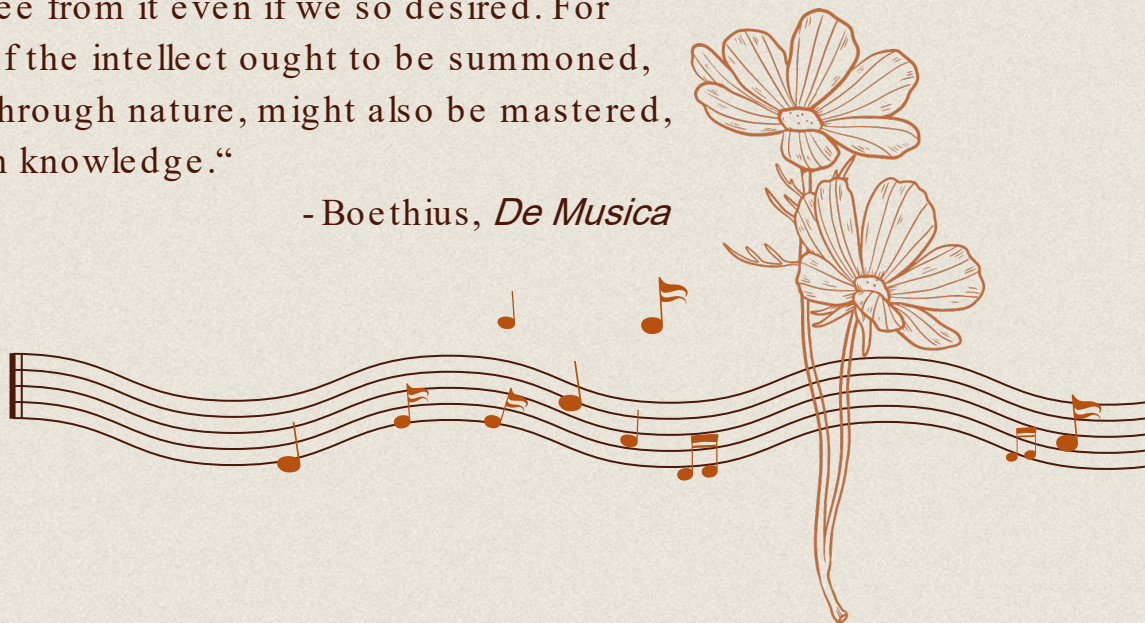
Music Theory Mod n

For AWM

"It appears without doubt that music is so naturally united with us that we cannot be free from it even if we so desired. For this reason the power of the intellect ought to be summoned, so that this art, innate through nature, might also be mastered, comprehended through knowledge."

- Boethius, *De Musica*

Anthony Christiana
George Washington





01

Music of the Spheres

How math informs music informs math
informs music informs math informs
music informs....



Pythagoras (c. 570 – c. 495 BC)

Def: (Acoustics) The science that deals with the production, control, transmission, reception, and effects of sound.

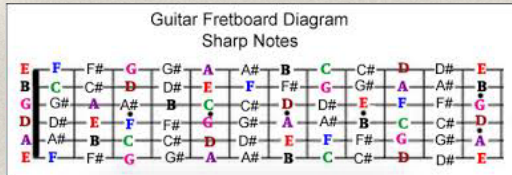
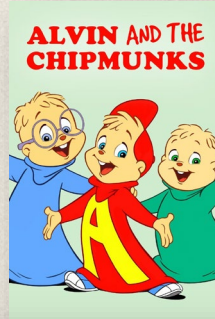
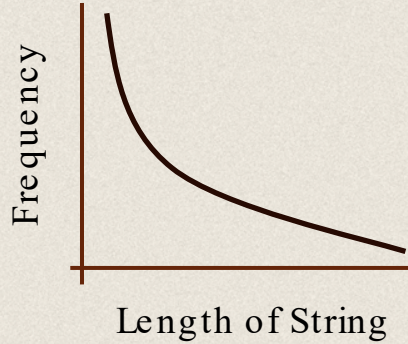
Experiment: Length and Pitch

Pluck two strings with lengths $l_1 > l_2$





Pythagoras (c. 570 - c. 495 BC)



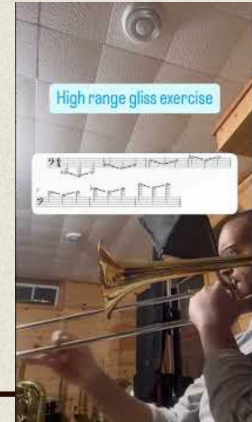
Pitch: Frequency (Hz) of sound waves

$$f = \frac{v}{2L}$$

Frequency: f

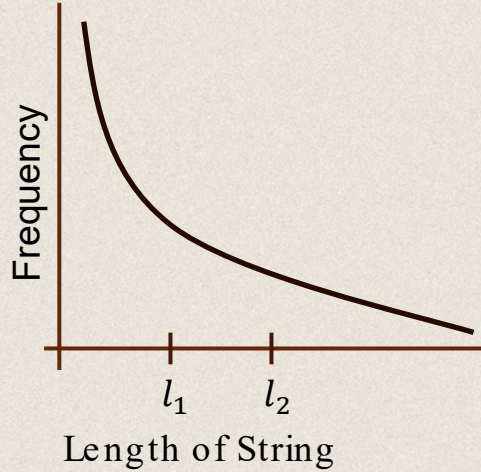
Length: L

Wave Speed: $v = \sqrt{T/\mu}$





Experiment:



$$f(l_1) > f(l_2)$$

Def: (Harmony) the combination of simultaneous musical notes



Question: What notes sound good together?

$$f(l_1)/f(l_2)$$

Which Do You Prefer?

l_1

l_2



$$\frac{f(l_1)}{f(l_2)} = \frac{3}{2}$$

l_1

l_2



$$\frac{f(l_1)}{f(l_2)} = \frac{64}{45}$$

l_1

l_2



$$\frac{f(l_1)}{f(l_2)} = \frac{21}{10}$$

l_1

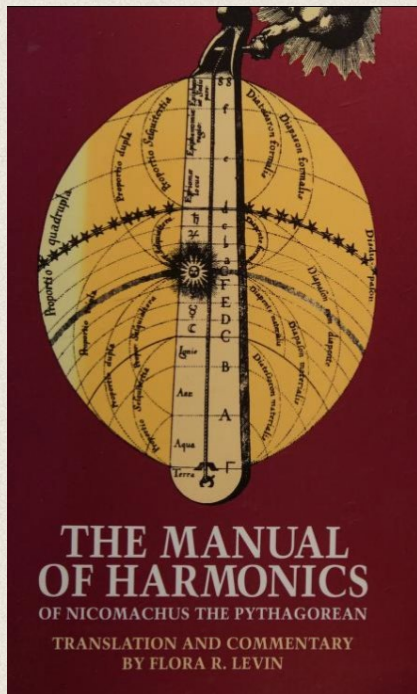
l_2



$$\frac{f(l_1)}{f(l_2)} = \frac{2}{1}$$



Pythagoras's influence lasted for over 2000 years. The illustration below is from a manuscript on the theory of music dated 1492.



Pythagoras (c. 570 - c. 495 BC)

Conclusion:

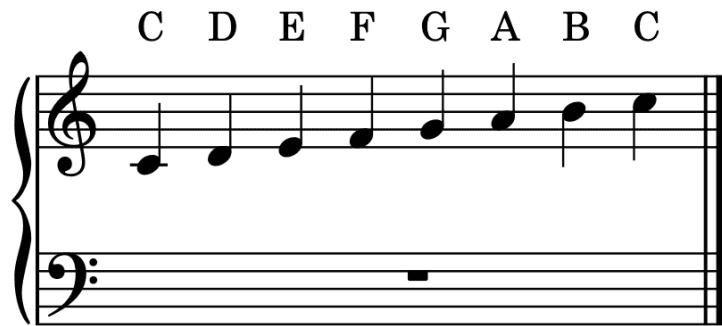
Consonances arise from ratios of small whole numbers.

$$\frac{f(l_1)}{f(l_2)} = \frac{m}{n} \quad \text{for } m, n \in \mathbb{N}$$

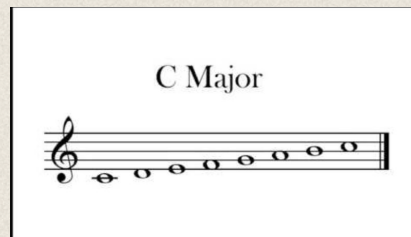
Note	Frequency ratio
C	1:1
D	9:8
E	5:4
F	4:3
G	3:2
A	5:3
B	15:8
C	2:1

Pythagoras (c. 570 – c. 495 BC)

Conclusion:
Consonances (nice harmony) arise from ratios of small whole numbers.



$$\frac{f(l_1)}{f(l_2)} = \frac{m}{n} \quad \text{for } m, n \in \mathbb{N}$$



Conclusion:

Consonances arise from ratios of small whole numbers.

Note	Frequency ratio
C	1:1
D	9:8
E	5:4
F	4:3
G	3:2
A	5:3
B	15:8
C	2:1

Q: Why does harmony work this way?

Does math determine aesthetic value?

Is it cultural?

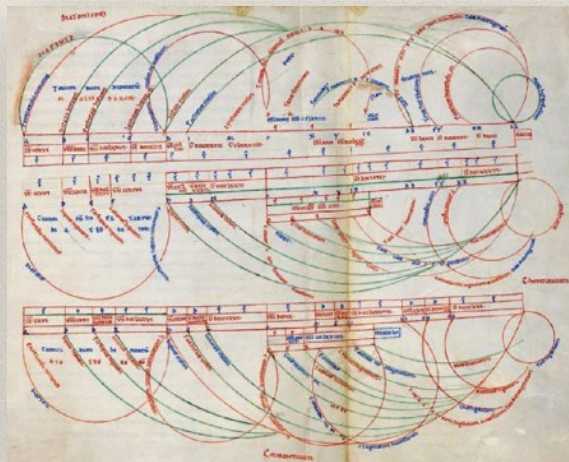
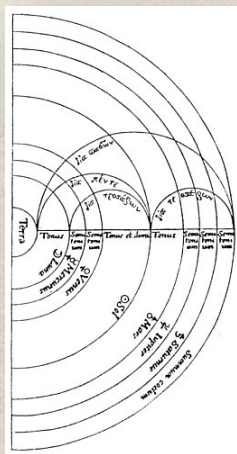
Is it evolutionary?

For Pythagoras, the connection between math and music was evidence of the divine, and this connection between math and music opened him up to some magical thinking...

Taking it too far: *Musica mundana*

Music of the Spheres:

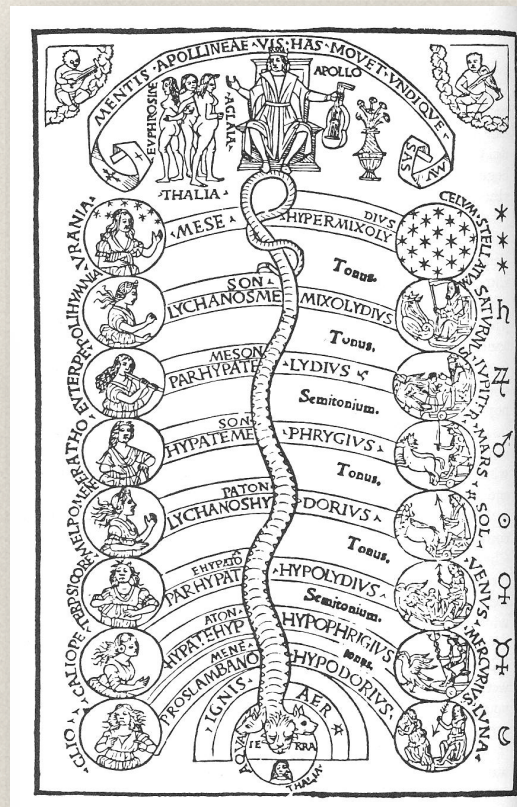
- The planets make a sound as they pass each other
- Ratios match up with harmonic ratios
- We can't hear this music because we are too used to it



(a)

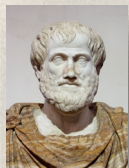


(b)



Musica mundana – Music of the Spheres

Aristotle, *On the Heavens*



" [Pythagoras] suppose[d] that the **motion of bodies of that size must produce a noise** , since on our earth the motion of bodies far inferior in size and in speed of movement has that effect. Also, when the sun and the moon, they say, and all the stars, so great in number and in size, are moving with so rapid a motion, **how should they not produce a sound immensely great** ? Starting from this argument and from the observation that their speeds, as measured by their distances, are in the **same ratios as musical concordances** , they assert that the **sound given forth by the circular movement of the stars is a harmony** .



Since, however, it appears unaccountable that we should not hear this music, they explain this by saying that the sound is in our ears from the very moment of birth and is thus indistinguishable from its contrary silence, since sound and silence are discriminated by mutual contrast."

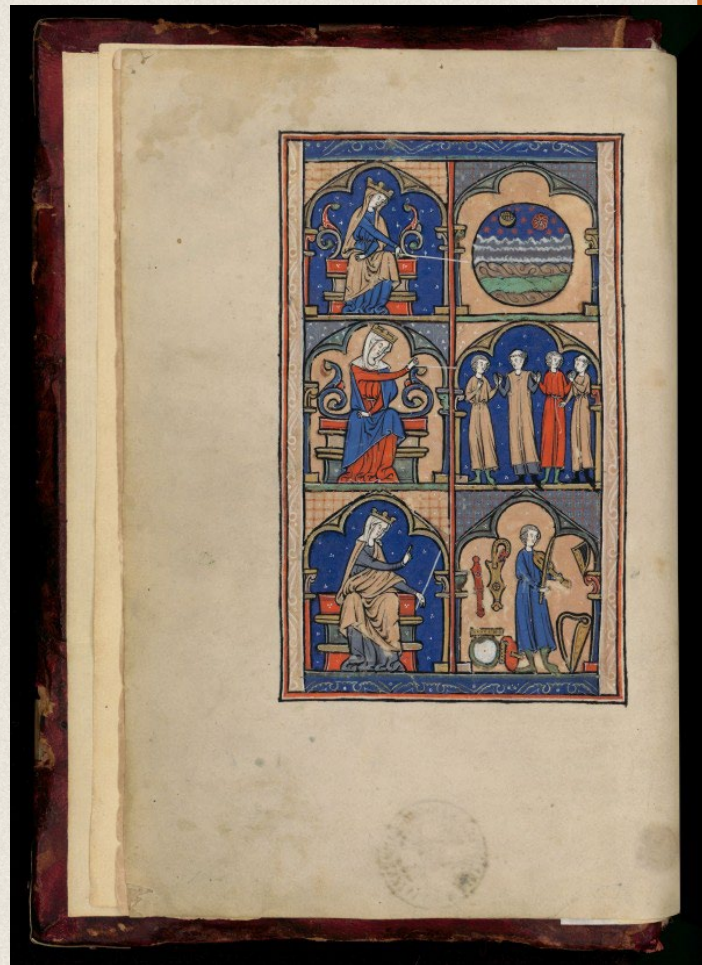


Boethius (c. 480–520)



Hierarchy of Music:

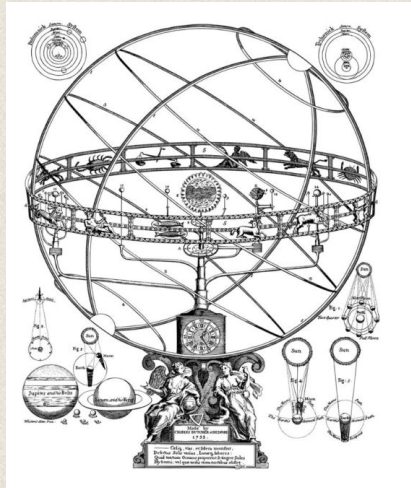
- *Musica Mundana* – cosmic harmony from the motion of the planets and the rhythm of the seasons
- *Musica Humana* – the harmony of the body and soul
- *Musica Instrumentalis* – singing and instrument playing



Kepler (1571 - 1630)



Earth's maximum and minimum speeds, for example, are in a ratio of roughly 16 to 15, or that of a semitone, whereas Venus' orbit is nearly circular, and therefore only produces a singular note. Mercury, which has the largest eccentricity, has the largest interval, a minor tenth, or a ratio of 12 to 5. This range, as well as the relative speeds between the planets, led Kepler to conclude that the Solar System was composed of two basses (Saturn and Jupiter), a tenor (Mars), two altos (Venus and Earth), and a soprano (Mercury), which had sung in "perfect concord," at the beginning of time, and could potentially arrange themselves to do so again.



HARMONICIS LIB. V. 207

omnia (infinita in potentiâ) permeantes actu : id quod aliter à me non potuit exprimi, quam per continuam seriem Notarum intermedia-

Saturnus Jupiter Mars ferè Terra

Venus Mercurius Hic locum habet etiam

The image shows a page from a book titled 'HARMONICIS LIB. V. 207'. It contains a Latin inscription and musical notation for the planets. The notation is arranged in two rows. The first row shows Saturnus, Jupiter, Mars ferè, and Terra. The second row shows Venus, Mercurius, and Hic locum habet etiam. The notation consists of notes on staves, with some notes being diamond-shaped and others being circles. The staves are labeled with the names of the planets.

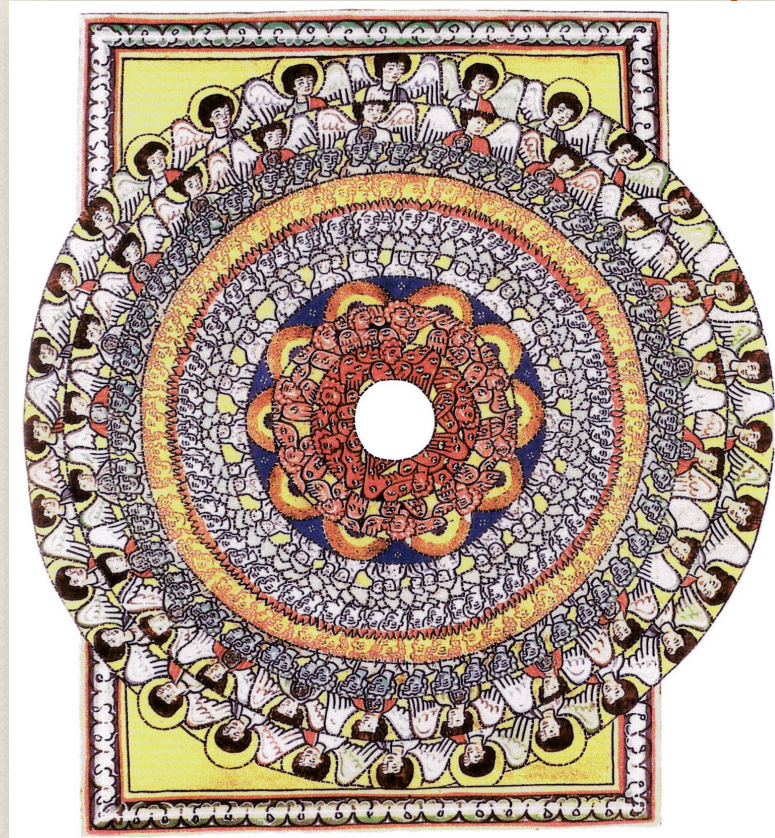
Hildegard von Bingen (c. 1098 – 1179)



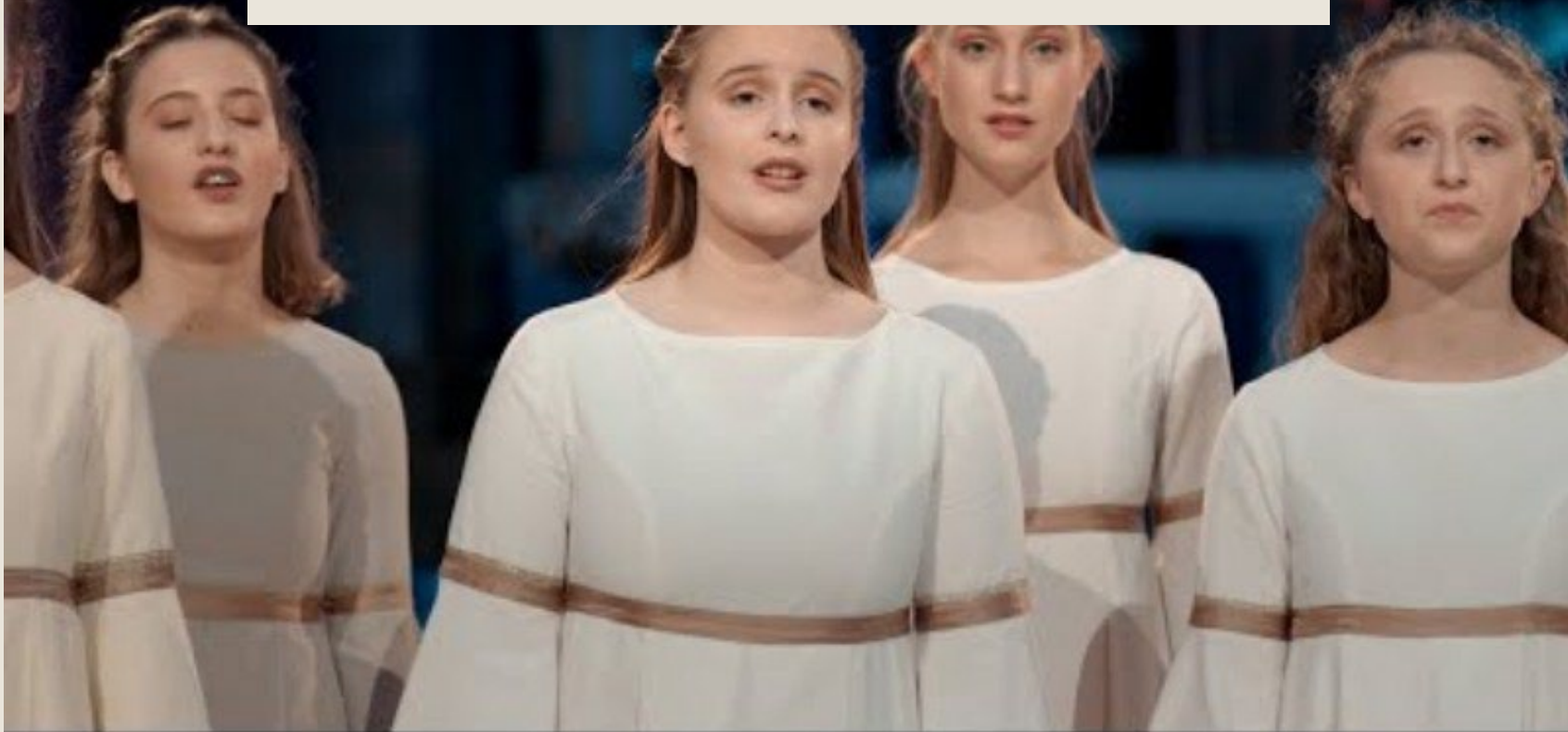
"Then I saw in the secret places in the heights of Heaven two armies of heavenly spirits who shone with great brightness...all these armies, as you hear, are singing with marvelous voices all kinds of music about the wonders that God works in blessed souls, by which God is magnificently glorified."



She believed that the lower musics were an attempt to mimic the divine music of the spheres, thus glorifying God.



Hildegard Von Bingen: De Spiritu Sancto





02

Sonification

Music of the Spheres(?)



Sounds of the Sun

Real life music
of the
spheres!



4 MIN READ

Sounds of the Sun

Stylized illustration of the Sun with an animation of a sound wave superimposed.
Credits: NASA's Goddard Space Flight Center

 **Jessica Evans**
Public Affairs Officer

Micheala M. Sosby

JUL 25, 2018

 ARTICLE

Data from ESA (European Space Agency) and NASA's Solar and Heliospheric Observatory (SOHO) has captured the dynamic movement of the Sun's atmosphere for over 20 years. Today, we can hear the Sun's movement — all of its waves, loops and eruptions — with our own ears.

This sound helps scientists study what can't be observed with the naked eye.

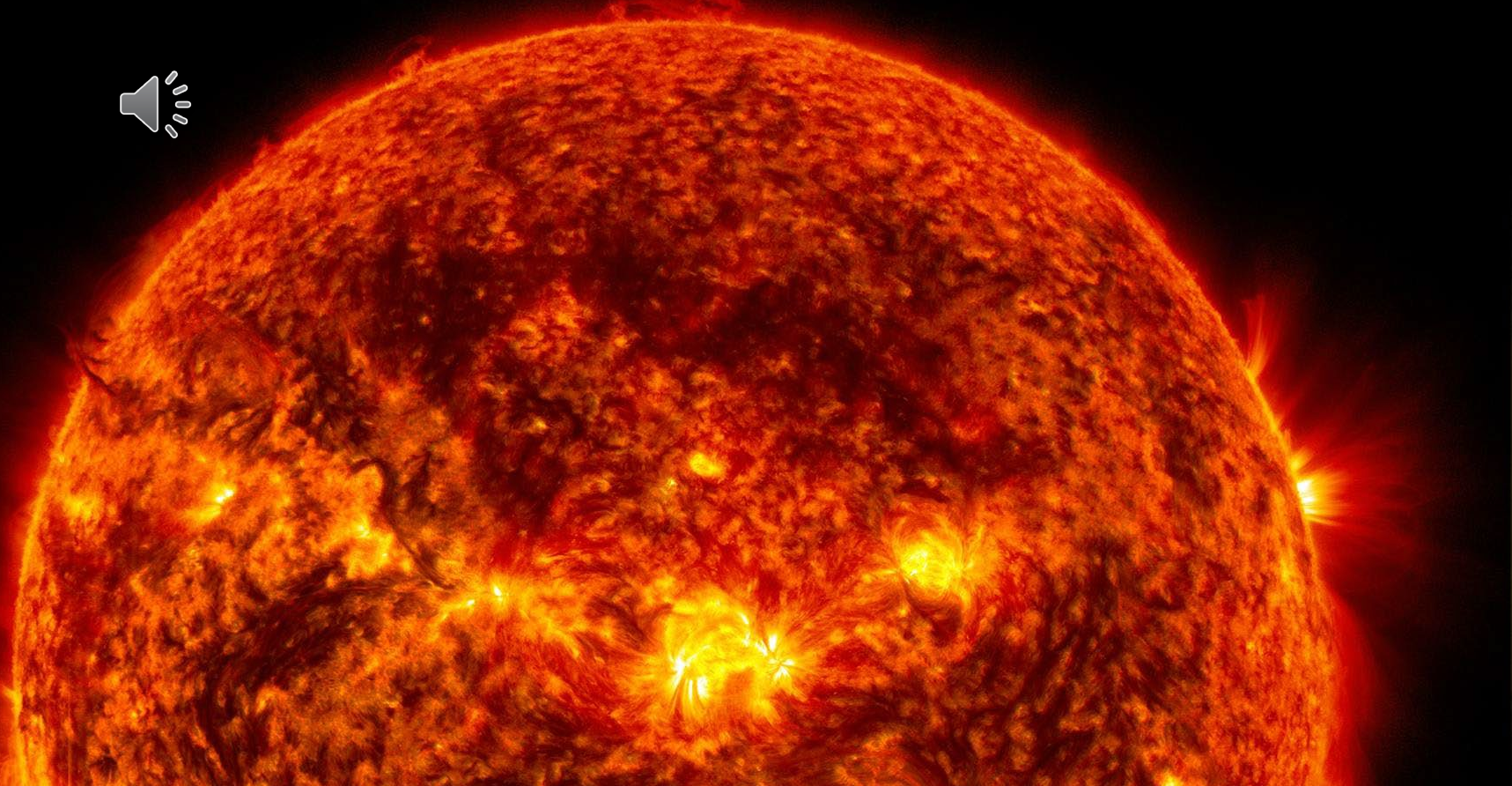
"Waves are traveling and bouncing around inside the Sun, and if your eyes were sensitive enough they could actually see this," said Alex Young, associate director for science in the Heliophysics Science Division at NASA's Goddard Space Flight Center in Greenbelt, Maryland.

Explore more music & audio like
Sounds of the Sun
on SoundCloud.





Sounds of the Sun, NASA



Sounds of the Sun, Me



SPACE IS A VACUUM

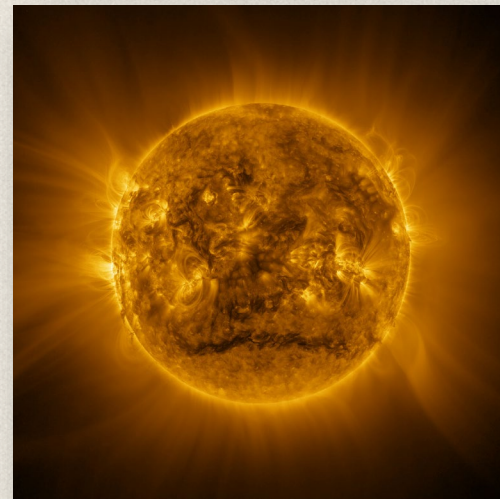
Def: (Sound) a mechanical disturbance from a state of equilibrium that propagates through an elastic material medium.

“Sonification” of the Sun

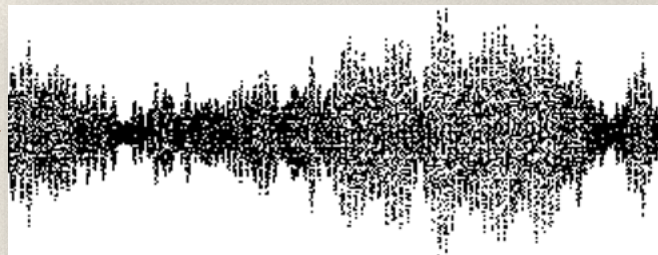
Def: (Sonification) The use of non - speech audio to represent data. (Analogue of “visualization”)

Sounds of the Sun

- Capture velocity data from waves of matter flowing around inside the sun
- Clean the data of junk to produce nice waves
- Multiply the frequencies by 42,000
- Voila! Science Communication.



Michelson Doppler Image data





SONIFICATION OF ARP 140

WIND OF BOSTON
DATA PAINTINGS





03

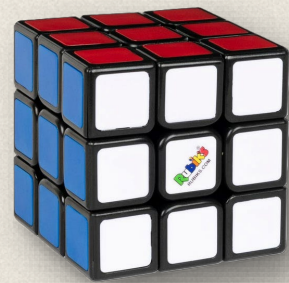
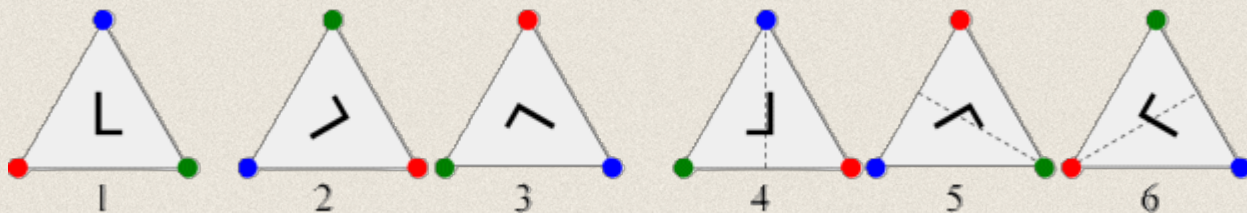
Music Theory Mod n

Group Theory

Loosely, a group G is an algebraic structure which describes the **symmetry** of an object/set.

Def: (Group) A group is a set G with a binary operation such that

1. (Identity) $\forall g \in G, \exists e \in G$ such that $g e = e g = g$
2. (Inverses) $\forall g \in G, \exists h \in G$ such that $gh = hg = e$
3. (Associativity) $\forall g, h, k \in G, (gh)k = g(hk)$



Cyclic Abelian Groups: $\mathbb{Z}_n$

Def: (Cyclic Group)

Fix n ; $\mathbb{Z}_n$ is the group where

- **Set:** $\mathbb{Z}_n = \{[0], [1], [2], \dots, [n-1]\}$
- **Operation:** “Clockwork arithmetic”

$$[a] + [b] = [c] \text{ where } c \text{ is the remainder of } (a + b)/n$$

Example: $n = 12$

$$[3] + [4] = [7]$$

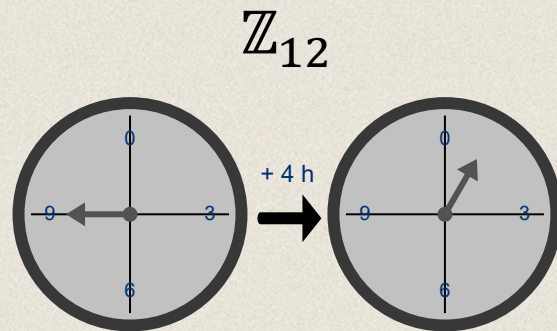
$$[0] + [a] = [a]$$

$$[8] + [6] = [2]$$

$$[7] + [5] = [0]$$

Identity: $[0]$

Inverse: $[a]$ has inverse $[n - a]$



Scales $\leftrightarrow \mathbb{Z}_n$

Def: (Scale) $\mathbb{Z}_n$

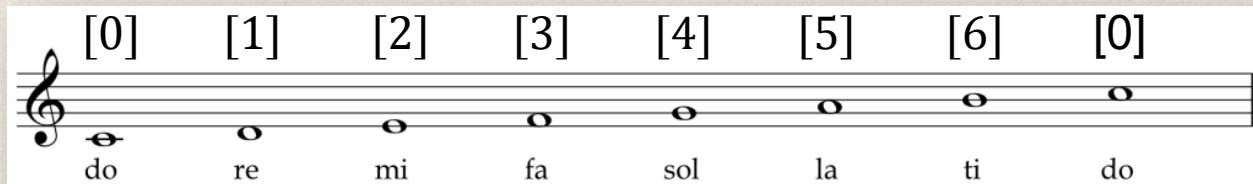
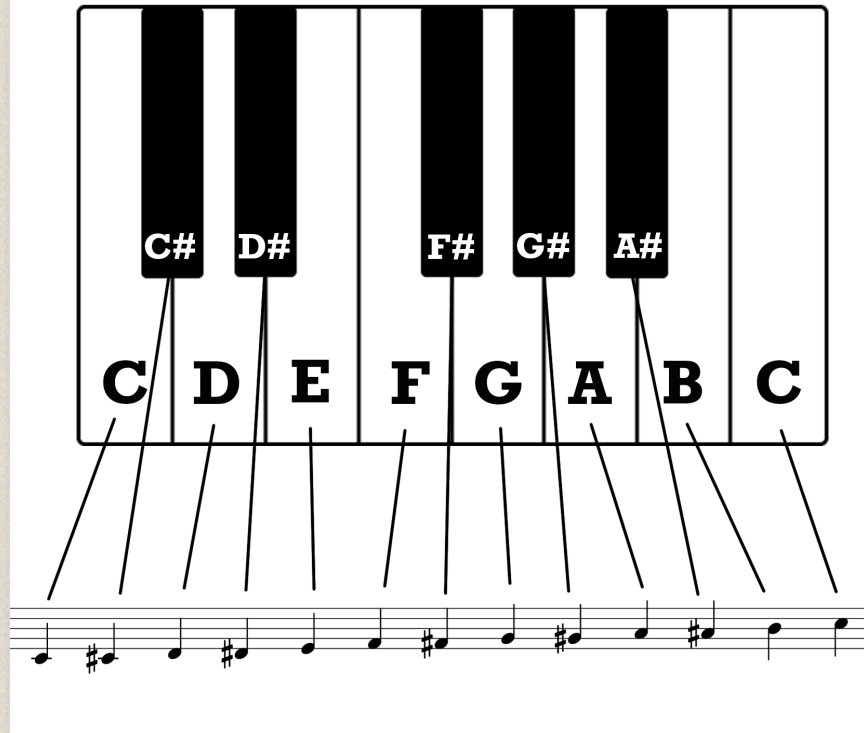
An increasing sequence of pitch classes.

Def: (Chromatic Scale) $\mathbb{Z}_{12}$

All 12 pitch classes in succession.

Def: (Diatonic Scale) $\mathbb{Z}_7$

A seven-note scale, e.g. the major scale.

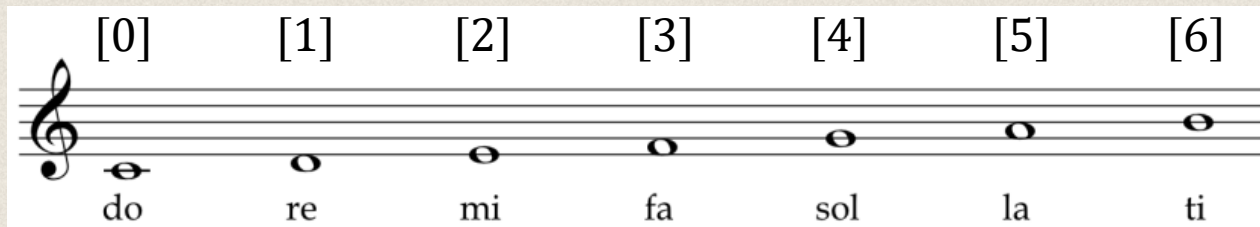


Diatonic Scales $\leftrightarrow \mathbb{Z}_7$

Def: (Diatonic Scale)

Any seven successive notes of different pitch classes

With the group homomorphism $\mathbb{Z} \rightarrow \mathbb{Z}_n$, we can sonify finite sequences in $\mathbb{Z}$.



[0]	[1]	[2]	[3]	[4]	[5]	[6]
do	re	mi	fa	sol	la	ti



Homomorphism

Music: $\mathbb{Z} \rightarrow \mathbb{Z}_n$

- Fix n .
- Identify $\mathbb{Z}_n$ with an n -note scale
- Build a sequence of integers
 $(a_n) = a_1, a_2, a_3, \dots$ for $a_i \in \mathbb{Z}$
- Listen to image of sequence under quotient map

$$p: \mathbb{Z} \rightarrow \mathbb{Z}_n$$

$$a_i \rightarrow [a_i]$$

Example: (Fibonacci Sequence)

- Fix $n = 12$
- $\mathbb{Z}_{12} = \{[0], [1], [2], [3], \dots, [11]\}$
 $= \{C, C\#, D, D\#, E, F, F\#, G, G\#, A, A\#, B\}$

Chromatic C scale.

- Fibonacci Sequence $f_1 = [0], f_2 = [1]$

$$f_n = f_{n-1} + f_{n-2}$$

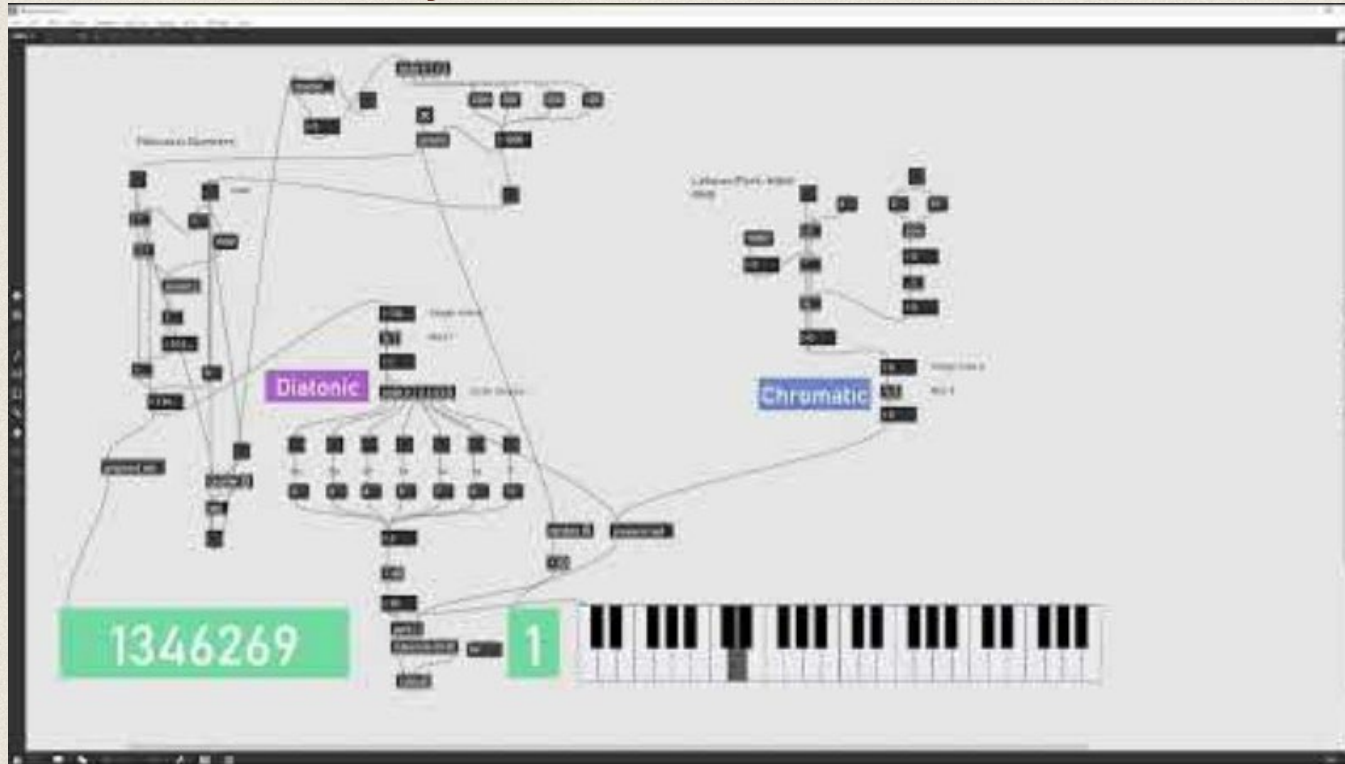
Listen to image of sequence under quotient

map

$$p: \mathbb{Z} \rightarrow \mathbb{Z}_n$$

$$a_i \rightarrow [a_i]$$

Sonification: Fibonacci Sequence Mod 7 (Diatonic Major)



(Psuedo)-Random Music

Def: (Lehmer/Park-Miller Random Number Generator)

- Fix p large prime
- Pick “multiplier” a of high multiplicative order mod p
- Pick “seed” X_0 coprime to p
- Build the sequence $X_{k+1} = [a \cdot X_k] = a \cdot X_k \pmod{p}$

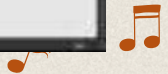
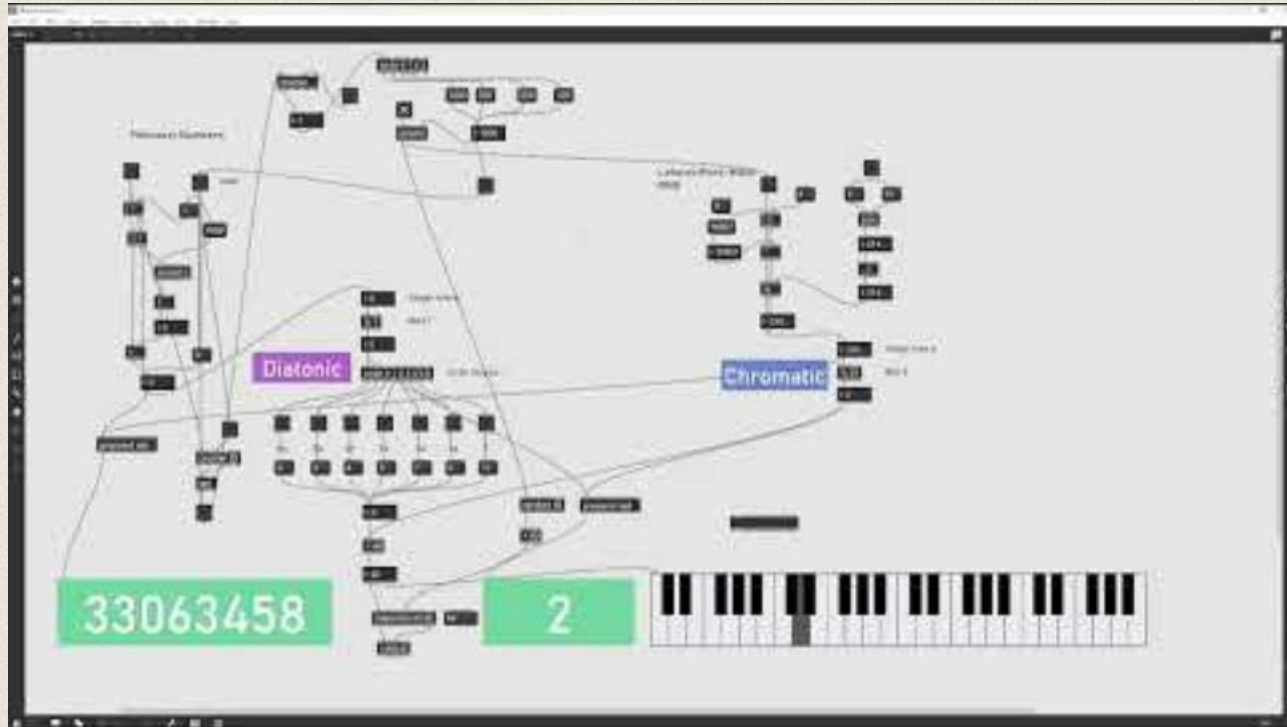
The X_k are pseudo-random numbers, used e.g. in C++

Example:

- Fix $p = 2^{31} - 1$
- Pick “multiplier” $a = 7^{35} = 16,807$
- Pick “seed” $X_0 = 2$
- Build the sequence $X_{k+1} \in \mathbb{Z}_p$ and send to the chromatic scale $\mathbb{Z}_{12}$



Sonification: L/PM RNG Music (Chromatic 2-Octave)



Would a composer
ever do this?



Serialism



Schoenberg



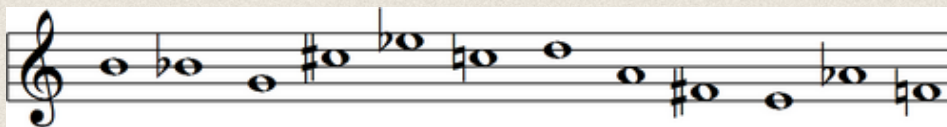
Berg



Webern

Def: (Serialism)

- Between 1912-1922, Schoenberg was looking for a new unifying philosophy for composition.
- In *tonal* music, there are scales which determine a *key*.
- He proposed a type of music called *12-Tone Serialism* without key centers: all 12 notes chromatic pitches are treated with equal importance.

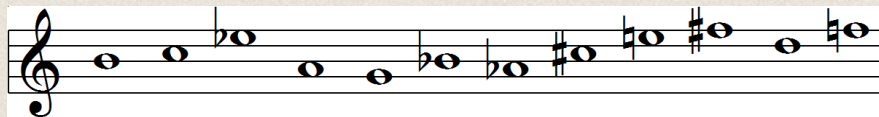


$$R = \{ [11], [10], [7], [1], [3], [0], [2], [9], [6], [4], [8], [5] \}$$



Rules for 12-Tone Composition

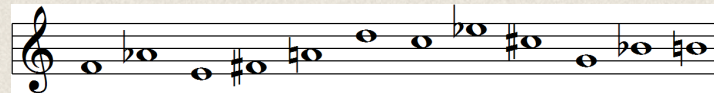
1. Pick a 12-note sequence $P = a_1, a_2, \dots, a_{12} \in \mathbb{Z}_{12}$ such that $a_i \neq a_j$ for $i \neq j$, called “the row”
2. All successive notes in your piece are
 1. The row P , possibly transposed: $P_k = (a_1 + k, a_2 + k, \dots, a_{12} + k)$



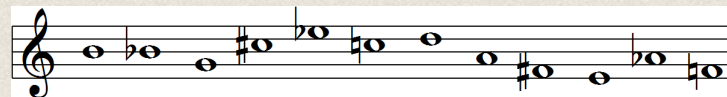
$P = \{ [11], [10], [7], [1], [3], [0], [2], [9], [6], [4], [8], [5] \}$

2. Images of P_k under the interval-preserving transformations of P :

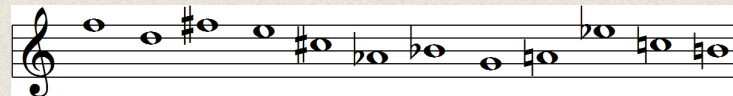
Inversion: $I = (-a_1, -a_2, \dots, -a_{12})$



Retrograde: $R = (a_{12}, a_{11}, \dots, a_1)$



Retrograde Inversion: $R \circ I = I \circ R$



И

ДЕТСКАЯ ПЬЕСА

(1924)

Liebllich¹⁾



527 528 529

530 531 532

This image shows a page of a musical score for a string quartet, specifically measures 527 through 532. The score is written for four staves, each representing a different string instrument: Violin I (top), Violin II, Viola, and Cello/Double Bass (bottom). The time signature is 4/4. The key signature has one flat (B-flat). The score includes various musical notations such as notes, rests, and dynamic markings like *pp* (pianissimo) and *mf* (mezzo-forte). There are also articulation marks like accents and slurs. The measures are numbered 527, 528, 529, 530, 531, and 532 at the top of each measure group. The notation is complex, with many beamed notes and rests, indicating a fast or intricate passage.





Elisabeth Lutyens
TRIO
Op. 57
(1964)



12-Tone Transformations as a Group Action

The group of interval-
preserving transformations

$\circ$	e	r	i	ri
e	e	r	i	ri
r	r	e	ri	i
i	i	ri	e	r
ri	ri	i	r	e

$$T \simeq \mathbb{Z}_{12}$$

$$T \cong \mathbb{Z}_2 \times \mathbb{Z}_2$$

Called the *Klein 4-Group*

- Music from group action on finite sets?
- Representation Theory music?



Aleatoric/Stochastic Music

From *alea*, Latin for “dice”.

Aleatoric music:

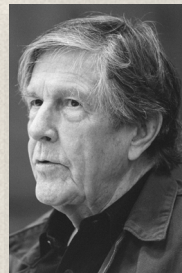
- In the mid 20th century composers like John Cage and Stockhausen wanted to make atonal music with fewer rules.
- They create aleatory pieces where some element of composition or performance is left to chance.

Stochastic music:

- Later, composers like Xenakis tried to open these “randomness” techniques even more.
- He looked to probability theory to create textures that mimicked the natural world
 - Ex: Free stochastic music, Markov music

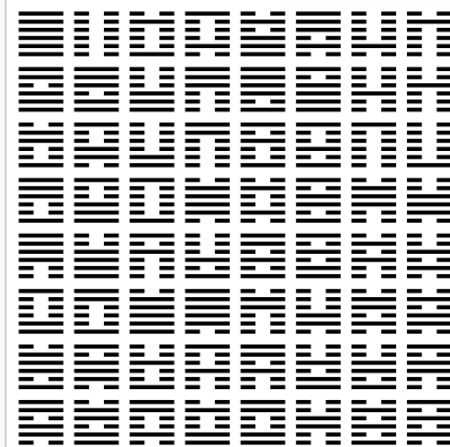
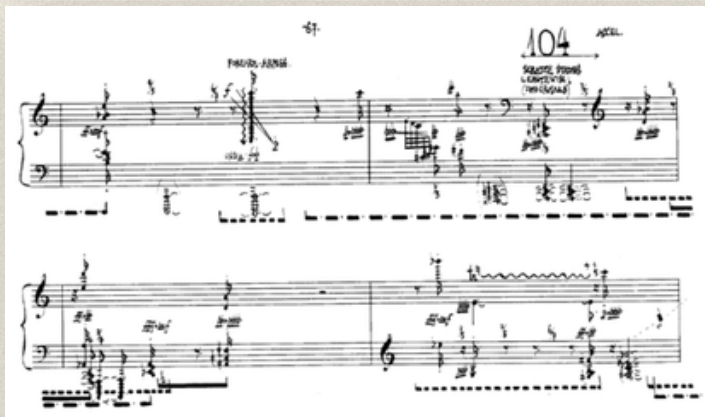


John Cage: *Music of Changes* (1951)



This piece uses I Ching or “Book of Changes”
to determine the performance.

Ancient divination text from (1000-750 BC)
from the Chinese Western Zhou period



I Ching divination involves obtaining
a hexagram by [random generation](#)
(such as [tossing coins](#)), then reading
the chapter associated with that
hexagram.





Iannis Xenakis: *Pithoprakta*(1955)

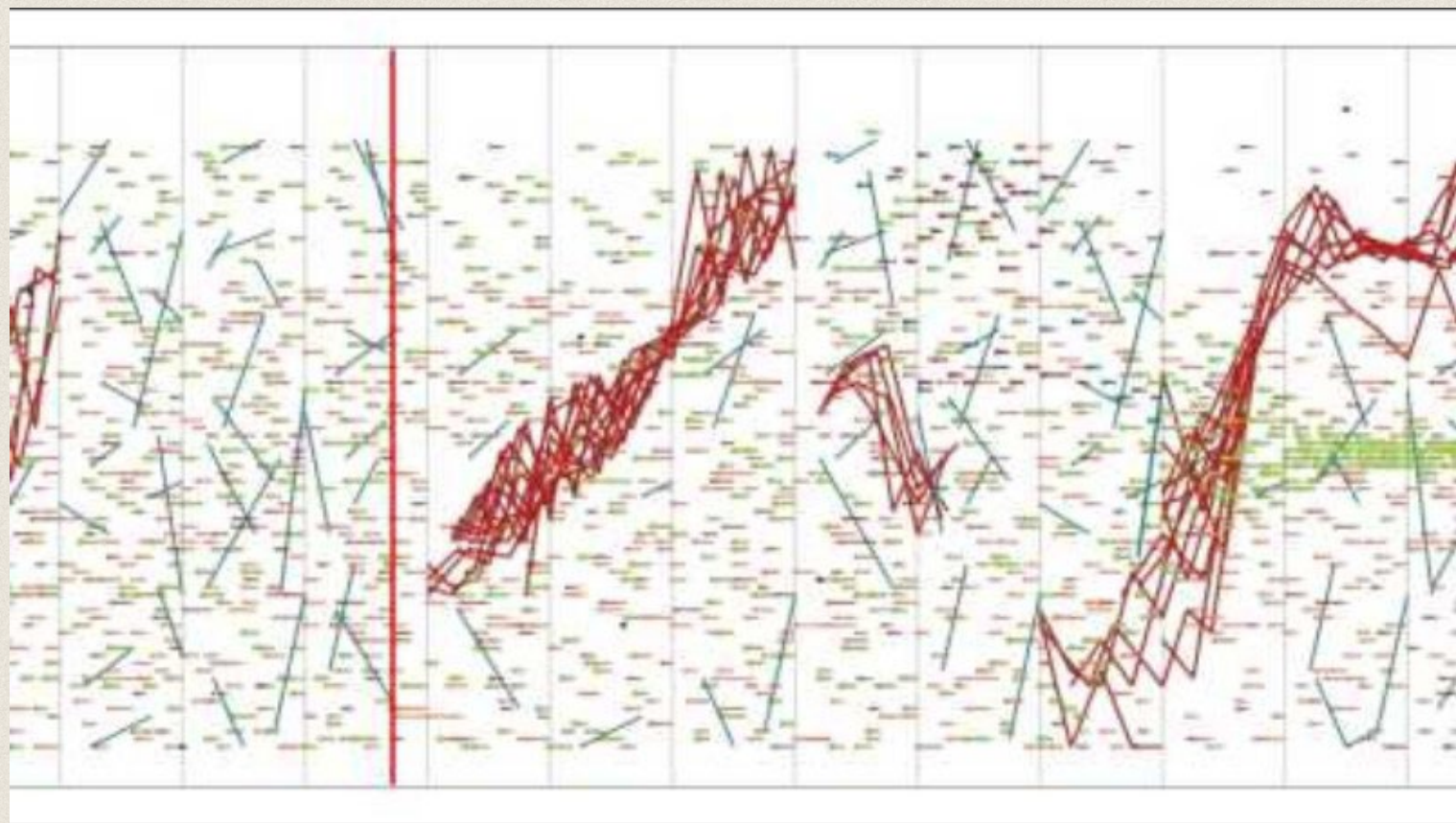
Pithoprakta; Greek for “actions through probability”

“This relates to Jacob Bernoulli's law of large numbers which states that as the number of occurrences of a chance event increases, the average outcome approaches a determinate end.

The piece is based on the statistical mechanics of gases,^[2] Gauss's law,^[3] or Brownian motion.^[4] Each instrument is conceived as a molecule obeying the Maxwell–Boltzmann distribution law,^[5] with Gaussian distribution of temperature fluctuation.”

-Wikipedia, *Pithoprakta*





Math and Music

- Math and Physics inform the base assumptions of our Western music theory practice

What “notes exist”? Why do they sound good?

- Sonification (Data $\rightarrow$ Music) as a tool for science communication or composition
- Abstract algebra as an alternative descriptive language for music theory
- Ideas in math---probability theory, group theory---as the impetus for composition



Thank you!

Contact me:

ajchristiana@gwmail.gwu.edu

